

```
In [1]: import pandas as pd
import numpy as np

import matplotlib.pyplot as plt
import seaborn as sns
```

```
In [6]: df = pd.read_csv(
    "superstore.csv",
    encoding='latin1'
)
```

View Dataset

```
In [7]: df.head()
```

```
Out[7]:
```

	Row ID	Order ID	Order Date	Ship Date	Ship Mode	Customer ID	Customer Name	Segment	Country
0	1	CA-2016-152156	11/8/2016	11/11/2016	Second Class	CG-12520	Claire Gute	Consumer	United States
1	2	CA-2016-152156	11/8/2016	11/11/2016	Second Class	CG-12520	Claire Gute	Consumer	United States
2	3	CA-2016-138688	6/12/2016	6/16/2016	Second Class	DV-13045	Darrin Van Huff	Corporate	United States
3	4	US-2015-108966	10/11/2015	10/18/2015	Standard Class	SO-20335	Sean O'Donnell	Consumer	United States
4	5	US-2015-108966	10/11/2015	10/18/2015	Standard Class	SO-20335	Sean O'Donnell	Consumer	United States

5 rows × 21 columns



Dataset Structure

Dataset Shape

```
In [8]: df.shape
```

```
Out[8]: (9994, 21)
```

Dataset Information

```
In [9]: df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 9994 entries, 0 to 9993
Data columns (total 21 columns):
#   Column                Non-Null Count  Dtype
---  -
0   Row ID                 9994 non-null   int64
1   Order ID               9994 non-null   object
2   Order Date             9994 non-null   object
3   Ship Date              9994 non-null   object
4   Ship Mode              9994 non-null   object
5   Customer ID            9994 non-null   object
6   Customer Name          9994 non-null   object
7   Segment                9994 non-null   object
8   Country                9994 non-null   object
9   City                   9994 non-null   object
10  State                  9994 non-null   object
11  Postal Code            9994 non-null   int64
12  Region                 9994 non-null   object
13  Product ID             9994 non-null   object
14  Category               9994 non-null   object
15  Sub-Category           9994 non-null   object
16  Product Name           9994 non-null   object
17  Sales                   9994 non-null   float64
18  Quantity               9994 non-null   int64
19  Discount               9994 non-null   float64
20  Profit                 9994 non-null   float64
dtypes: float64(3), int64(3), object(15)
memory usage: 1.6+ MB
```

Column Names

```
In [11]: df.columns
```

```
Out[11]: Index(['Row ID', 'Order ID', 'Order Date', 'Ship Date', 'Ship Mode',
                'Customer ID', 'Customer Name', 'Segment', 'Country', 'City', 'State',
                'Postal Code', 'Region', 'Product ID', 'Category', 'Sub-Category',
                'Product Name', 'Sales', 'Quantity', 'Discount', 'Profit'],
              dtype='object')
```

Check Missing Values

```
In [12]: df.isnull().sum()
```

```
Out[12]: Row ID          0
         Order ID       0
         Order Date     0
         Ship Date      0
         Ship Mode       0
         Customer ID    0
         Customer Name   0
         Segment        0
         Country        0
         City           0
         State          0
         Postal Code     0
         Region         0
         Product ID     0
         Category       0
         Sub-Category    0
         Product Name    0
         Sales           0
         Quantity       0
         Discount        0
         Profit          0
         dtype: int64
```

Remove Duplicate Rows

```
In [13]: df.duplicated().sum()
```

```
Out[13]: 0
```

Convert Date Column

```
In [14]: df['Order Date'] = pd.to_datetime(
         df['Order Date']
         )
```

Monthly Revenue Trends

Calculate Monthly Revenue

```
In [17]: df['Month'] = df['Order Date'].dt.to_period('M')
```

```
In [18]: monthly_sales = df.groupby(
         'Month'
         )['Sales'].sum()

         print(monthly_sales.head())
```

```
Month
2014-01    14236.895
2014-02     4519.892
2014-03    55691.009
2014-04    28295.345
2014-05    23648.287
Freq: M, Name: Sales, dtype: float64
```

Plot Monthly Revenue Trend

```
In [19]: plt.figure(figsize=(12,6))

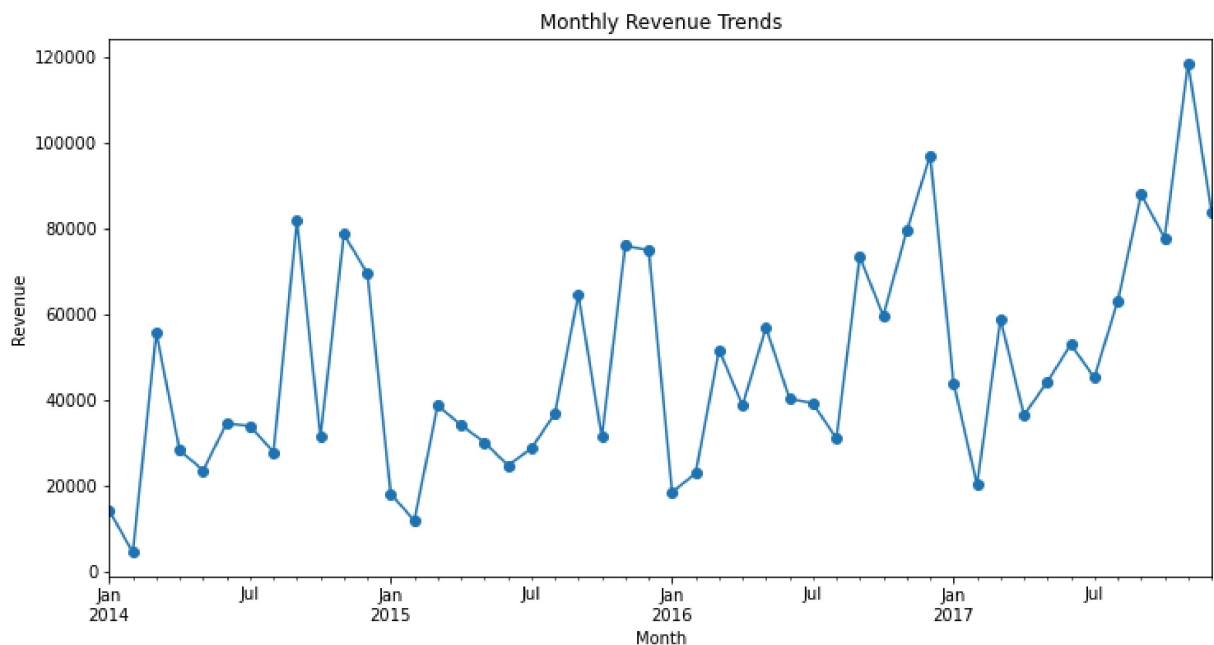
monthly_sales.plot(
    marker='o'
)

plt.title(
    "Monthly Revenue Trends"
)

plt.xlabel("Month")

plt.ylabel("Revenue")

plt.show()
```



Top-Selling Products

```
In [20]: top_products = df.groupby(
    'Product Name'
)['Sales'].sum()\
    .sort_values(
```

```
ascending=False
).head(10)
```

In [21]: `print(top_products)`

```
Product Name
Canon imageCLASS 2200 Advanced Copier      6159
9.824
Fellowes PB500 Electric Punch Plastic Comb Binding Machine with Manual Bind  2745
3.384
Cisco TelePresence System EX90 Videoconferencing Unit      2263
8.480
HON 5400 Series Task Chairs for Big and Tall      2187
0.576
GBC DocuBind TL300 Electric Binding System      1982
3.479
GBC Ibimaster 500 Manual ProClick Binding System      1902
4.500
Hewlett Packard LaserJet 3310 Copier      1883
9.686
HP Designjet T520 Inkjet Large Format Printer - 24" Color      1837
4.895
GBC DocuBind P400 Electric Binding System      1796
5.068
High Speed Automatic Electric Letter Opener      1703
0.312
Name: Sales, dtype: float64
```

Visualization

In [22]: `plt.figure(figsize=(12,6))`

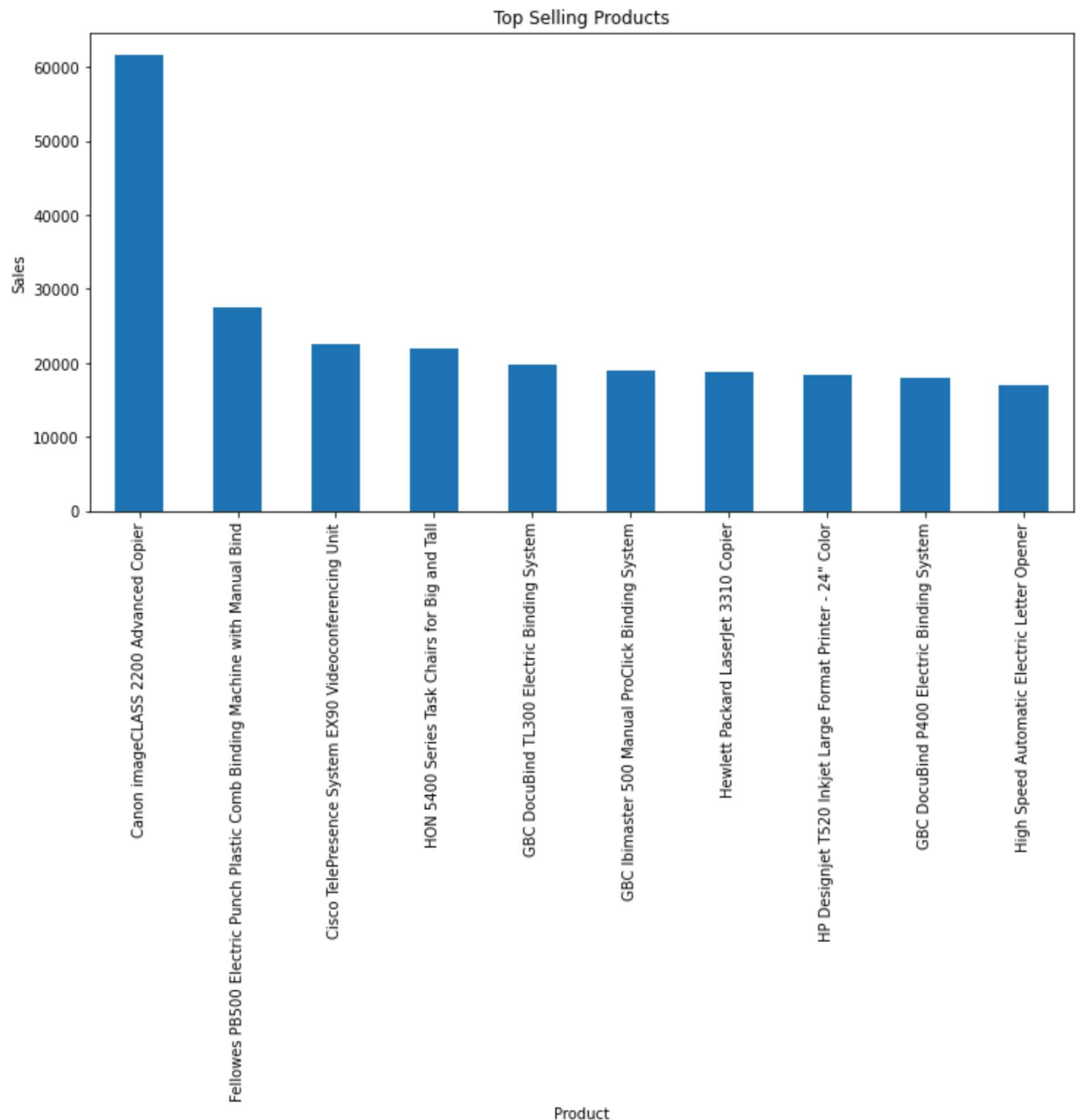
```
top_products.plot(
    kind='bar'
)

plt.title(
    "Top Selling Products"
)

plt.xlabel("Product")

plt.ylabel("Sales")

plt.show()
```



Profit Analysis

Total Profit

```
In [23]: total_profit = df[
         'Profit'
         ].sum()

         print(total_profit)
```

286397.0217

Profit by Category

```
In [24]: profit_category = df.groupby(  
        'Category'  
        )['Profit'].sum()
```

```
In [25]: print(profit_category)
```

```
Category  
Furniture      18451.2728  
Office Supplies 122490.8008  
Technology     145454.9481  
Name: Profit, dtype: float64
```

Visualization

```
In [27]: plt.figure(figsize=(8,5))  
  
profit_category.plot(  
    kind='bar',  
    color='green'  
    )  
  
plt.title(  
    "Profit by Category"  
    )  
  
plt.xlabel("Category")  
  
plt.ylabel("Profit")  
  
plt.show()
```



Sales vs Profit Relationship

Scatter Plot

```
In [28]: plt.figure(figsize=(8,5))

sns.scatterplot(

x=df['Sales'],

y=df['Profit']

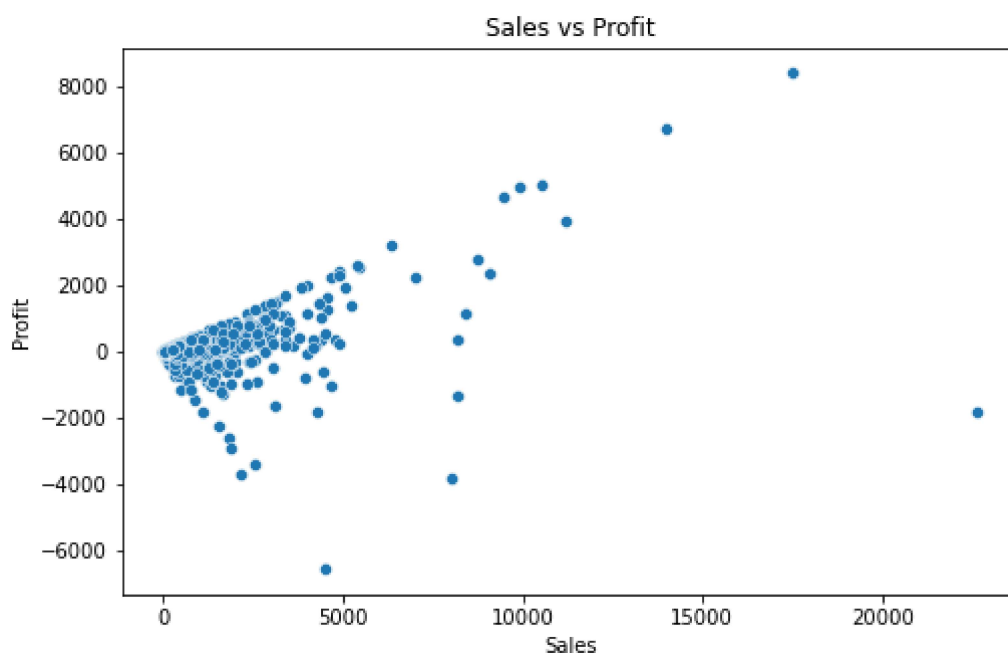
)

plt.title(

"Sales vs Profit"

)

plt.show()
```



Category-wise Sales Analysis

Calculate Category Sales

```
In [29]: category_sales = df.groupby(

'Category'

)['Sales'].sum()
```



```
In [33]: plt.figure(figsize=(8,5))

sns.barplot(

x=category_sales.index,

y=category_sales.values

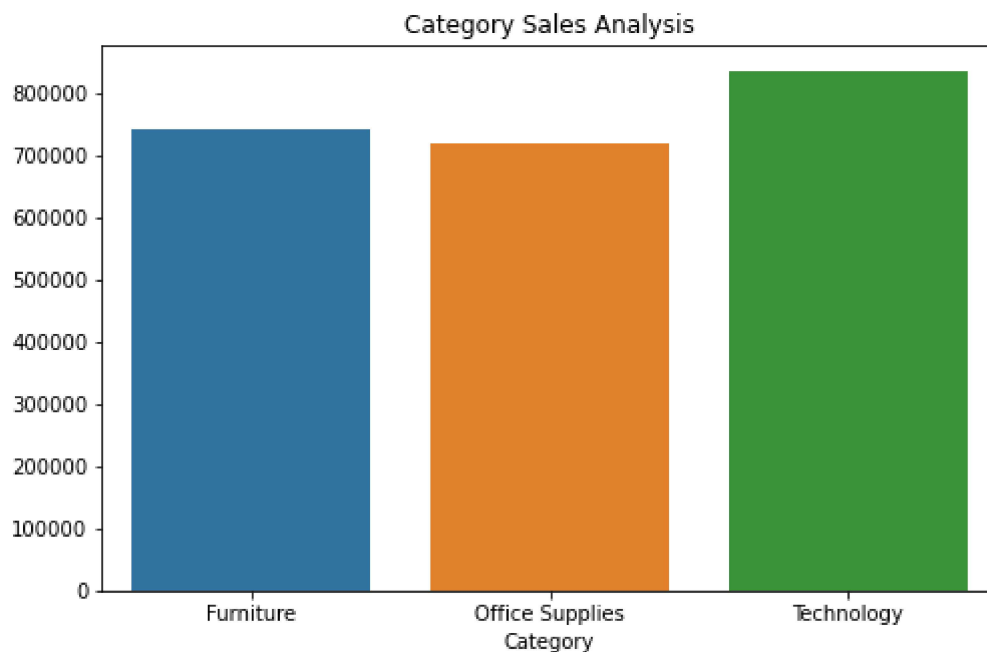
)

plt.title(

"Category Sales Analysis"

)

plt.show()
```



Insight

Insight 1

Monthly revenue changes over time and shows seasonal business variation.

Insight 2

Top products contribute a large share of overall sales.

Insight 3

Some categories generate strong sales but lower profit.

Insight 4

Sales and profit do not always increase together.

Conclusion

The sales data analysis provided important insights into business performance.

Monthly revenue analysis highlighted revenue trends over time.

Top-selling products were identified based on total sales contribution.

Profit analysis revealed differences across product categories and showed that higher sales do not always lead to higher profitability.

These insights can support better business decision-making and strategy planning.

In []: